# Blockchain & Secure Multi-Party Computation

Master 2 — Computer Science

Session 2 of 12 — Introduction to Blockchain II

## Where We Are

Session 1 established what a blockchain is, its five fundamental principles, and its history. This session completes the introduction by covering the four types of blockchain (public, private, consortium, and hybrid) and exploring the wide range of sectors where blockchain technology creates real value. Understanding both the types and the applications helps in choosing the right blockchain model for a given problem.

## 1 Types of Blockchain

Not every blockchain is open to the public, and not every blockchain needs to be. The four types differ along three main dimensions: who can read the ledger, who can write (submit transactions), and who can participate in validating (consensus).

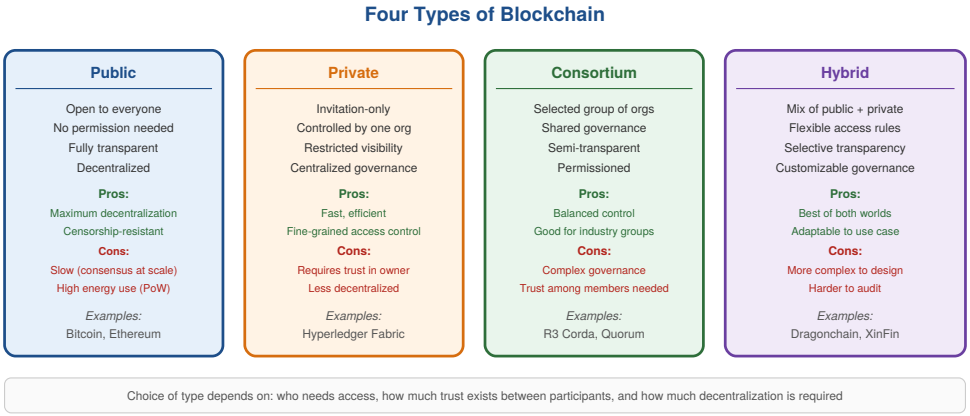



Figure 1: The four types of blockchain differ in access control, governance, and the trust assumptions they make about participants.

### 1.1 Public Blockchain

**Definition — Public Blockchain**

A public blockchain is a fully open network where any person or machine can: (1) read all transaction history, (2) submit transactions, and (3) participate in consensus (validation). No permission is required from any authority. The network is governed by the protocol itself, with rules enforced by code and economic incentives.

**Example — Bitcoin and Ethereum**

Bitcoin and Ethereum are the most prominent public blockchains. Anyone worldwide can download the software, run a node, mine (or stake), and transact. The consensus mechanism

(Proof of Work in Bitcoin, Proof of Stake in Ethereum post-2022) ensures that the network agrees on a single ledger without any central coordinator. This makes them resistant to censorship, single points of failure, and government shutdown.

**? Think about it**

Why does a public blockchain need an economic incentive (like mining rewards)?

**✓ Answer**

Because validating transactions and running a full node consumes real resources (electricity, hardware, bandwidth). Without a reward, rational actors would not volunteer these resources. The mining reward (newly created Bitcoin) and transaction fees create a financial incentive: honest miners earn rewards, while cheating costs resources with no reward. This aligns the economic interest of validators with the security of the network.

**? Think about it**

Mining Reward vs. Transaction Fees

**✓ Answer**

The total compensation awarded to the successful miner per block is governed by:

$$\text{Total Block Reward} = \text{Block Subsidy (New BTC)} + \sum \text{Transaction Fees}$$

| Comparison Metric | Mining Reward | Transaction Fees |
|---|---|---|
| **Source of Funds** | Newly minted BTC created by the protocol | Existing BTC transferred from sender wallets |
| **Payer** | The Bitcoin protocol automatically via the `Coinbase` transaction | Network users broadcasting transactions |
| **Recipient** | The miner or mining pool validating the block | The miner or mining pool validating the block |
| **Value Determinant** | Programmatically fixed, halving every 210,000 blocks (currently 3.125 BTC) | Dynamic; determined by network congestion and block space demand (mempool) |
| **Primary Role** | Currency issuance, supply distribution, and bootstrapping security | Spam prevention, transaction prioritization, and long-term network security incentive |
| **Future Outlook** | Decreases to zero around the year 2140 (capped at 21 million BTC) | Permanent; will become the sole financial incentive for miners |

Table 1: Comparison between the Block Subsidy and Transaction Fees in Bitcoin.

## 1.2 Private Blockchain

**Definition — Private Blockchain**

A private blockchain is controlled by a single organization, which decides who may read the ledger, who may submit transactions, and who participates in consensus. Access is strictly controlled. The organization typically runs all (or most) validating nodes itself.

**Example — Hyperledger Fabric in enterprise**

Hyperledger Fabric (a Linux Foundation project) is a popular private blockchain framework used by enterprises for internal processes such as interbank settlements, supply chain tracking within a company, and audit trails for regulatory compliance. It offers fast transaction processing (thousands per second vs. Bitcoin's 7) because consensus among a few trusted internal nodes is much simpler than consensus among thousands of anonymous global nodes.

**? Think about it**

If a private blockchain is controlled by one organization, how is it different from a traditional database?

**✓ Answer**

A fair question. In many cases, it is not very different. The value a private blockchain adds over a traditional database is limited to: an immutable audit trail that even internal administrators cannot secretly modify (the hash chain still applies), a standardized data structure for sharing information with external partners on a need-to-know basis, and the ability to gradually transition toward more decentralized governance in the future. Critics argue that many private blockchain deployments would function equally well — or better — as a conventional database.

## 1.3 Consortium (Federated) Blockchain

**Definition — Consortium Blockchain**

A consortium blockchain is governed by a predefined group of organizations rather than a single entity. For example, 10 banks might each run one or more validating nodes and collectively decide the network's rules. Reading may be open to members only, or selectively public. Writing is restricted to authorized participants.

**Example — R3 Corda in banking**

R3 Corda is a distributed ledger platform designed specifically for financial institutions. A consortium of major banks (including HSBC, Barclays, and Deutsche Bank) uses it to settle interbank transactions directly, reducing the settlement time from days to seconds. Each bank runs a node and participates in validating transactions relevant to it — only the parties to a transaction see its details, protecting commercial privacy while maintaining verifiable settlement records.

**? Think about it**

Why would competing banks agree to share a blockchain consortium?

**✓ Answer**

Because the shared infrastructure benefits all of them. Today, when Bank A sends funds to Bank B, it goes through SWIFT and multiple correspondent banks, taking days and incurring fees. On a shared consortium blockchain, settlement is direct, near- instant, and cheaper for all participants — a case where competitors cooperate because the collective gain outweighs the competitive disadvantage of sharing infrastructure.

### 1.4 Hybrid Blockchain

**Definition — Hybrid Blockchain**

A hybrid blockchain combines elements of both public and private blockchains. Some data is kept private (visible only to authorized participants), while other data or records are made public and verifiable by anyone. The governance model can be custom- designed to fit specific needs.

**Example — Dragonchain and government records**

Dragonchain allows organizations to run private business logic and transaction data on internal chains, while publishing cryptographic proofs (hashes) of these transactions to a public blockchain (like Ethereum) for external verifiability. This lets a government agency, for example, keep citizen data private while still allowing anyone to verify that records have not been altered, using the public chain as a tamper-evident notary service.

**? Think about it**

Which type of blockchain is "best"?

**✓ Answer**

There is no universally best type. The choice depends entirely on the use case. Public blockchains are best when trust among parties is impossible or undesirable, and censorship resistance is critical. Private blockchains suit internal enterprise processes where speed and privacy matter more than decentralization. Consortium blockchains suit industry-wide infrastructure where multiple competing organizations need shared infrastructure. Hybrid blockchains suit cases requiring both privacy and public accountability.

## 2 Permissioned vs. Permissionless: A Clarification

The terms "public/private" and "permissioned/permissionless" are often confused. They describe different dimensions:

| | **Permissionless** | **Permissioned** |
|---|---|---|
| **Read access** | Anyone | Controlled by admin |
| **Write access** | Anyone | Controlled by admin |
| **Who validates** | Any node (incentivized) | Pre-approved nodes |
| **Typical type** | Public blockchain | Private or Consortium |
| **Example** | Bitcoin, Ethereum | Hyperledger, Corda |

Note: a public blockchain is always permissionless; a private blockchain is always permissioned. A consortium blockchain is permissioned but may have a public-readable ledger (partially permissioned).

# 3 Blockchain Applications and Use Cases

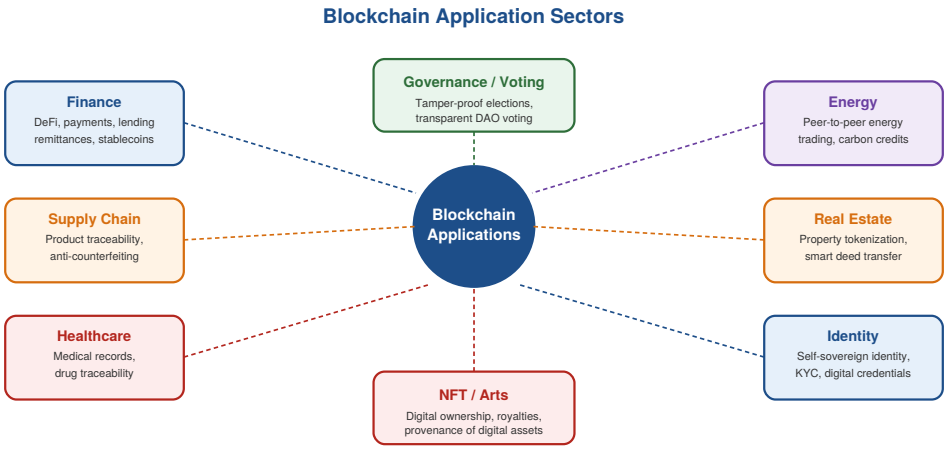



Figure 2: Blockchain technology has found application across many sectors. Each application benefits from at least one of blockchain's core properties: immutability, decentralization, transparency, or programmability (smart contracts).

## 3.1 Financial Services and Decentralized Finance

The most mature and widely deployed blockchain applications are in finance.

**Example — Remittances: sending money across borders**

A migrant worker in France wants to send 500 euros to family in Algeria. Using traditional banking, this takes 2–5 days and costs 5–10% in fees (SWIFT charges, correspondent banks, currency conversion). Using a blockchain-based remittance service (such as Stellar or Ripple), the same transfer completes in seconds for a fraction of a percent in fees. For the 281 million international migrants worldwide sending money home, this is a significant economic impact.

**Example — Decentralized Finance (DeFi)**

DeFi recreates traditional financial services (lending, borrowing, trading, derivatives, insurance) using smart contracts on public blockchains, without any bank, broker, or exchange as an intermediary. Users deposit cryptocurrency as collateral and borrow against it — the smart contract automatically liquidates the collateral if its value falls below the loan threshold. The entire process is governed by auditable code, not by a company's internal policies.

**? Think about it**

Give an example from the real world.

**✓ Answer**

Aave, a DeFi lending protocol on Ethereum, holds billions of dollars in user deposits and has issued billions in loans, all governed by smart contracts. There are no loan officers, no credit checks, and no human intervention — a user deposits Ethereum, borrows stablecoins against it, and repays at any time. The smart contract enforces all terms automatically, 24 hours a day, 7 days a week.

## 3.2 Supply Chain and Traceability

**Example — Food safety: tracing a contaminated product in seconds**

In 2018, a major E.coli outbreak in romaine lettuce in the United States took weeks to trace back to the source farm using paper records and phone calls. Walmart, after this incident, required its leafy green suppliers to record all supply chain steps on the IBM Food Trust blockchain (built on Hyperledger Fabric). Today, Walmart can trace the origin of any food product in its stores in seconds by scanning a QR code, rather than weeks. This could save lives in a food safety emergency.

**? Think about it**

Could a supplier simply enter false data into the blockchain?

**✓ Answer**

They could enter false data at the moment of input, but they cannot change it afterward. The blockchain does not validate the real-world truth of data at input time — this is called the "oracle problem." However, over time, repeated false entries would be contradicted by other participants' records, physical audits, or IoT sensor data (temperature sensors, GPS trackers) that record independently. The blockchain provides an immutable audit trail of what was recorded, even if it does not prevent initial false entry.

## 3.3 Healthcare

**Example — Patient-controlled medical records**

Today, a patient's medical history is fragmented across hospitals, clinics, and pharmacies that rarely share data securely. A blockchain-based health record system stores encrypted records on a distributed ledger where the patient holds the decryption key — they alone decide which doctor can read which portion of their history. This gives patients true ownership of their own data while maintaining an immutable, auditable record of who accessed what and when.

**Example — Drug traceability and counterfeit prevention**

The World Health Organization estimates that 10% of medicines in low-income countries are counterfeit. The MediLedger Network (a pharmaceutical consortium blockchain) records every step of a drug's journey from manufacturer to pharmacy. A pharmacist can scan a drug's QR code and instantly verify its complete chain of custody on the blockchain, detecting any break in the verified supply chain that might indicate counterfeiting.

## 3.4 Governance and Voting

**Example — Decentralized Autonomous Organizations (DAOs)**

A DAO is an organization whose governance rules are encoded in smart contracts on a public blockchain. Members hold governance tokens; proposals are voted on and automatically executed by the smart contract if they reach a threshold. There are no executives, no board, and no human administrator who could override the vote. MakerDAO, which governs the DAI stablecoin, has millions of dollars in treasury assets managed entirely through on-chain DAO governance.

**? Think about it**

Could blockchain be used for national elections?

**✓ Answer**

Technically possible, but practically complex. Blockchain provides verifiability (every voter can check their vote was counted), immutability (votes cannot be changed after submission), and transparency (anyone can audit results). However, challenges remain: ensuring voter identity without compromising anonymity, preventing coercion (if voting is verifiable by design, a coercer can demand proof of how you voted), and universal digital access. Estonia uses a blockchain-adjacent e-voting system for some elections, and several cities have piloted blockchain voting.

## 3.5 Digital Identity

**Example — Self-Sovereign Identity (SSI)**

Traditional digital identity is controlled by service providers (Facebook Login, government databases). SSI reverses this: a person holds their own identity credentials (education certificates, age verification, professional licenses) as cryptographically signed tokens on a blockchain, and shares only what is necessary with any verifier. Example: to prove you are over 18, you share only an age verification credential — not your name, address, or ID number. The verifier checks the credential's blockchain-anchored signature without contacting the issuer in real time.

## 3.6 Real Estate and Asset Tokenization

**Definition — Asset Tokenization**

Asset tokenization is the process of representing ownership of a real-world asset (property, artwork, a company's equity, commodities) as digital tokens on a blockchain. Each token represents a fraction of the asset. Tokenization enables fractional ownership, 24/7 trading, and automated dividend distribution through smart contracts.

**Example — Tokenizing a building**

A commercial building worth 10 million euros is tokenized into 10 million tokens of 1 euro each on a blockchain. Instead of needing 10 million euros to invest in commercial real estate (a barrier excluding most investors), any person worldwide can buy 100 tokens for 100 euros and own a proportional stake. Rental income is automatically distributed monthly by smart contract, proportional to token holdings. The RealT platform (operating on Ethereum) has been doing exactly this with US residential properties since 2019.

**? Think about it**

Give an example from the real world.

**✓ Answer**

The St. Regis Aspen Resort in Colorado (USA) tokenized 18.9% of its equity on the Ethereum blockchain, allowing accredited investors to buy tokens representing fractional ownership of the luxury hotel. This was one of the first SEC-compliant real estate security token offerings, raising $18 million.

## 3.7 Energy and Sustainability

**Example — Peer-to-peer energy trading**

In Brooklyn, New York, the Brooklyn Microgrid project allows neighbors with solar panels to sell excess electricity directly to neighbors who need it, using a blockchain to record and settle micro-transactions automatically, without the utility company as an intermediary. Each kilowatt-hour traded is recorded immutably, and smart contracts handle payment instantly. This model incentivizes local renewable energy production and could transform how energy is distributed in communities.

**Example — Carbon credits on blockchain**

Carbon credits (certificates representing one tonne of $CO_2$ saved or removed) have historically suffered from double-counting: the same credit sold to two buyers, or credits issued for projects that never materialized. Blockchain provides a single, immutable registry where each credit is a unique token. Once "retired" (used to offset emissions), it is permanently burned and can never be reused. Projects like Toucan Protocol and KlimaDAO are building this infrastructure on Ethereum.

**? Think about it**

Why is blockchain particularly well-suited to carbon credit markets?

**✓ Answer**

Because carbon credits have exactly the properties blockchain addresses best: they are intangible, they need to be unique (no double-counting), they need a verifiable chain of custody from issuance to retirement, and they involve multiple mutually distrustful parties (project developers, verifiers, buyers, regulators) who currently rely on centralized registries that have proven susceptible to fraud. Blockchain provides a tamper-evident, globally accessible, and auditable alternative.

## 3.8 Comparing Blockchain Use Cases

It is useful to map each use case against the specific blockchain property it primarily exploits:

| **Use Case** | **Immutability** | **Decentralization** | **Transparency** | **Smart Contracts** |
|---|---|---|---|---|
| DeFi / Finance | Medium | High | High | High |
| Supply chain | High | Medium | High | Medium |
| Healthcare records | High | Medium | Low (private) | Low |
| Voting | High | High | Medium | High |
| Digital identity | High | High | Low (selective) | Medium |
| NFT / Ownership | High | Medium | High | High |
| Real estate tokens | High | Medium | Medium | High |
| Carbon credits | High | Medium | High | Medium |

### 3.9 Non-Fungible Tokens (NFTs) and Digital Ownership

**Definition — Non-Fungible Token (NFT)**

An NFT is a unique digital token recorded on a blockchain, representing ownership of a specific digital (or physical) asset. Unlike Bitcoin (where each unit is identical and interchangeable — "fungible"), each NFT has a unique identifier. The blockchain record establishes who owns the token and provides a public, verifiable provenance history.

**Example — NFTs beyond digital art**

While NFTs became famous for digital art sales, their deeper value is in provenance and programmable royalties. A musician can sell a song as an NFT with a smart contract clause that automatically pays the artist 10% every time the NFT is resold — something impossible to enforce in traditional music licensing. In gaming, in-game items as NFTs allow players to truly own and trade assets across games, outside the control of any game developer.

**? Think about it**

If an NFT is "just a token on the blockchain," does buying an NFT of a digital image give you the copyright to that image?

**✓ Answer**

Not automatically. An NFT is a record of ownership of the token — not necessarily of the intellectual property the token references. Most NFT marketplaces only transfer the token itself. Unless the smart contract or accompanying legal agreement explicitly transfers the copyright, the original creator retains it. This distinction between owning a token and owning the underlying IP is a common source of confusion and ongoing legal debate.

This "trustless trust" is the unique value proposition that distinguishes blockchain from any centrally administered alternative.

## 4 When Not to Use a Blockchain

A balanced understanding requires recognizing when blockchain is the wrong tool. Common mistakes include using blockchain simply because it is fashionable, or assuming every problem that involves multiple parties needs it.

**Example — Situations where a traditional database is better**

A hospital network wants to share patient records among five affiliated clinics that are all operated by the same healthcare group. There is already a legal and organizational trust relationship between the clinics. The data is highly sensitive and must be modifiable (diagnoses change, prescriptions are updated). In this case, a well-secured traditional database with role-based access control (Session 6 of the IoT Security module) is faster, easier to maintain, easier to correct, and just as secure as a blockchain — without the consensus overhead, energy cost, or immutability that would actually make the system harder to operate.

A useful rule summarized by practitioners is: "You probably do not need a blockchain if: (1) you can name the trusted party who should control the database; (2) all participants already have an existing legal relationship that creates accountability; or (3) modifying records is a normal, expected part of the system's operation."

**? Think about it**

A university wants to issue digital diplomas that employers can verify independently. Should it use a public blockchain, a private blockchain, or a traditional database?

**✓ Answer**

A public blockchain (or at minimum, a public notarization service anchored to one) is the right choice here — specifically because the verifying party (the employer) has no existing relationship with the university's IT infrastructure and cannot be expected to trust a private database the university controls. By anchoring diploma hashes on a public blockchain, the employer verifies the diploma's authenticity independently, without needing to call the university or trust its servers. This is exactly the "trustless verification" use case blockchain is designed for. The actual diploma content is typically stored off-chain (to protect student privacy), with only a hash stored on-chain.

## Session Summary

This session covered the four blockchain types (public, private, consortium, and hybrid) along with the key distinction between permissioned and permissionless networks. It then surveyed blockchain applications across finance (DeFi, remittances), supply chain (Walmart Food Trust), healthcare (patient records, drug traceability), governance (DAOs, voting), digital identity (SSI), digital ownership (NFTs), real estate tokenization, and energy trading and carbon credits. The comparison table mapped each use case to the specific blockchain property it primarily exploits.

The common thread across all these applications is trust between parties who would not naturally trust each other. Blockchain provides a neutral, rule-based, mathematically enforced infrastructure that no participant controls but all can independently verify. Not every problem requires a blockchain: if participants already trust each other, if only one organization is involved, or if speed and simplicity matter more than transparency and immutability, a traditional database remains the better engineering choice. Session 3 examines the cryptographic primitives — hash functions, digital signatures, and encryption — that make the blockchain's core guarantees possible.

**? Think about it**

After reviewing all these use cases, what single test should an engineer apply before choosing blockchain over a traditional database?

**✓ Answer**

"Do multiple mutually distrustful parties need to share a data record, and is there no neutral central authority acceptable to all of them?" If the answer is yes, and if immutability or auditability of that record matters, blockchain is a strong candidate. If the answer is no — if one organization controls the data or all parties already trust a common administrator — a traditional database is faster, simpler, cheaper, and easier to modify when requirements change.